

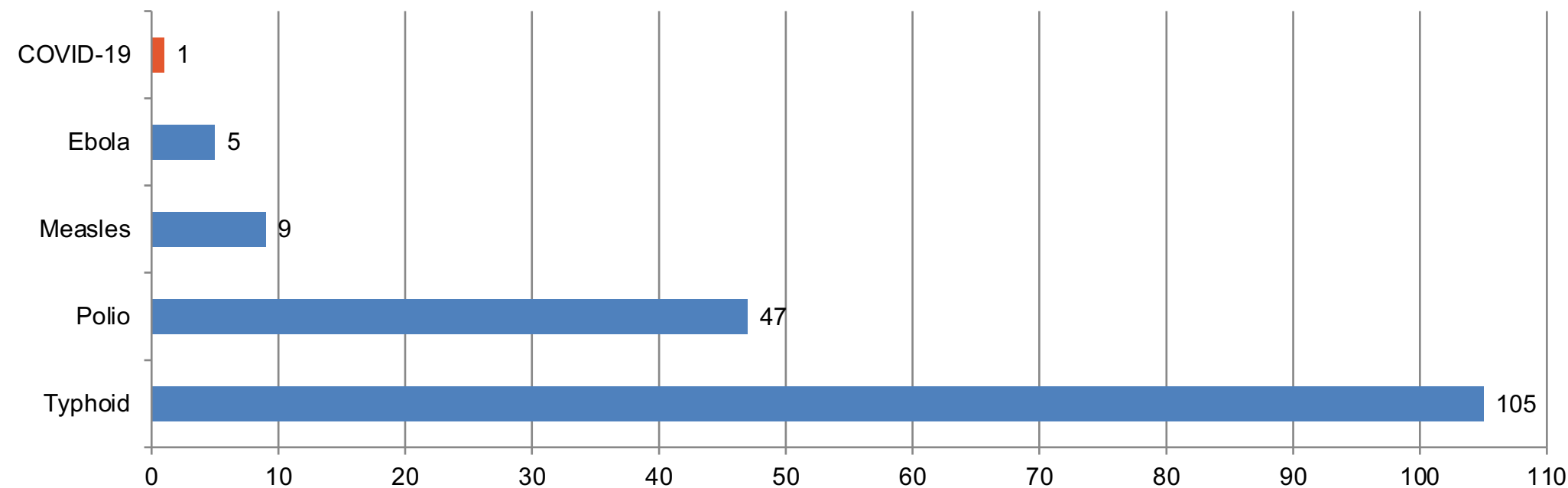
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of vaccination impact and innovation (1796–2020)

Audience: Public health professionals and policy analysts

Source attribution: Our World in Data (as compiled in source brief)

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



Time from pathogen identification to first vaccine licensure (years)

COVID-19 emphasized: <1 year (record timeline).

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Diphtheria	~21,000 cases / year	0 cases	100%
Polio (paralytic)	~16,000 cases / year	0 cases	100%
Smallpox	~48,000 cases / year	0 cases	100%
Pneumococcal disease	Baseline reference in brief	Residual burden remains	40.5% cases, 31.3% deaths
Hepatitis A	Baseline reference in brief	Residual burden remains	89% case reduction
Acute Hepatitis B	Baseline reference in brief	Residual burden remains	83.9% case reduction

Diseases at 100% are fully eliminated in US surveillance data from brief.

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Heatmap-style reference visualization (1929–2022)
State-level measles incidence transitions from high intensity pre-1963
to near-zero levels after 1980, with sporadic localized outbreaks.

First measles vaccine introduced

MMR vaccine introduced

US kindergarten vaccination
mandate

1963

1971

1980

Smallpox: The Only Human Disease Eradicated Through Vaccination

1959

— WHO resolution to eradicate smallpox

1967

— Intensified global eradication campaign launched

1977

— Last natural case reported (Somalia)

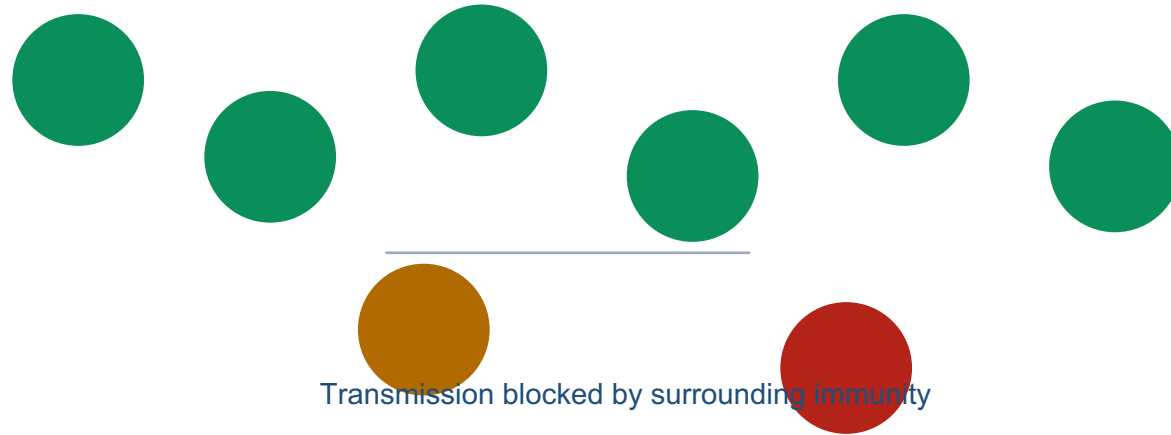
1980

— Global eradication certified

Why eradication succeeded

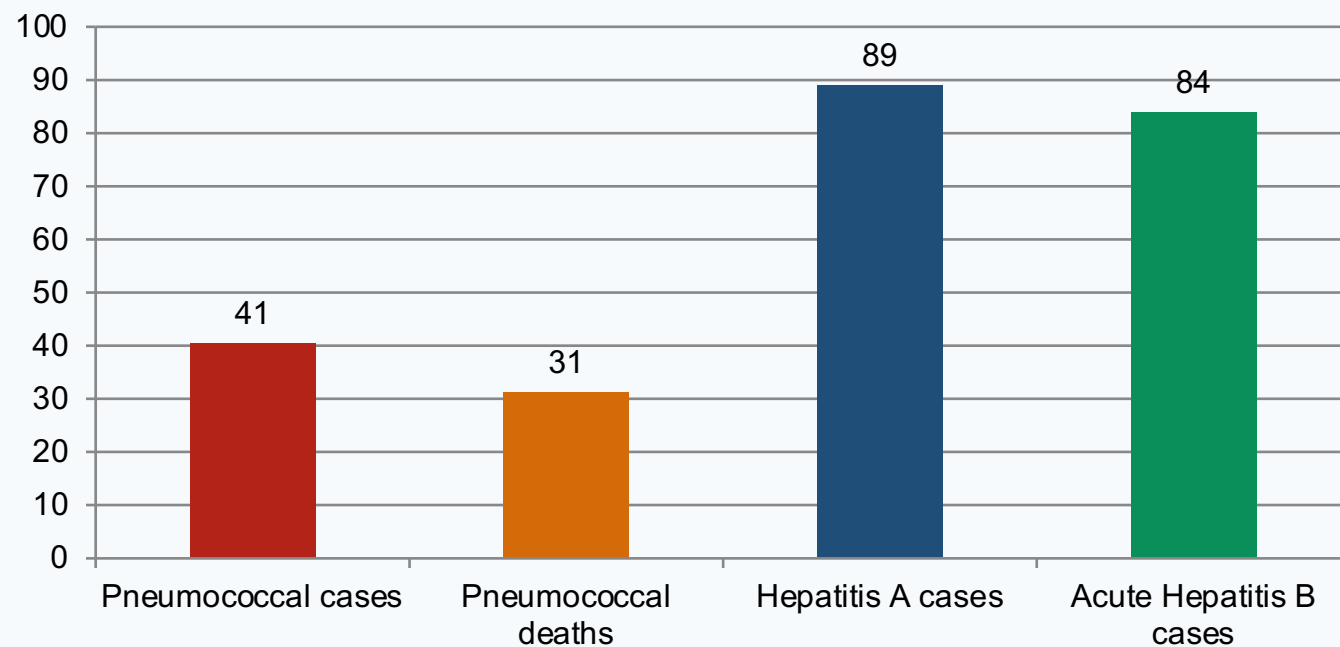
- Virus infected only humans
- Symptoms were distinguishable
- Vaccine was cheap and effective

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Vaccination at scale interrupts chains of infection, protecting those who cannot be vaccinated: newborns, immunocompromised individuals, and patients on chemotherapy.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



Barriers to closing gaps

- Unequal access to vaccines
- Supply shortages
- Limited trust and scientific understanding

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.
- Vaccine development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19).
- Measles incidence fell to near zero across all 50 states after the 1980 kindergarten mandate.
- Smallpox remains the only human disease eradicated globally, certified in 1980.
- Coverage gaps persist for pneumococcal disease, hepatitis A, and hepatitis B; health systems must improve access, coverage, and trust.

Call to action: Invest in equitable vaccine delivery, resilient supply chains, and community trust to translate scientific progress into universal protection.